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Going Deeper into Regression Analysis with Assumptions, Plots & Solutions

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Introduction

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All models are wrong, but some are useful – George Box

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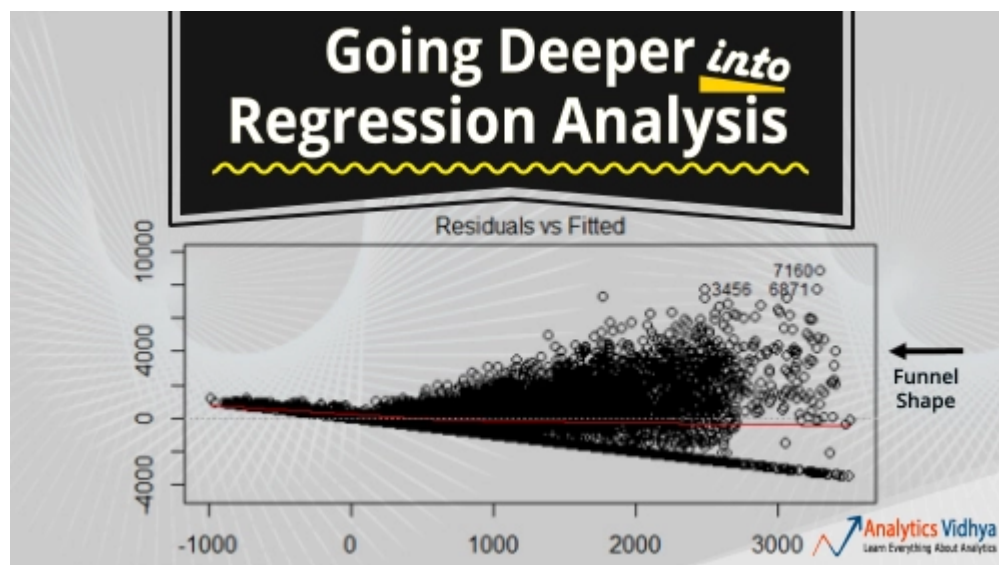
Regression analysis marks the first step in predictive modeling. No doubt, it's fairly easy to implement. Neither it's syntax nor its parameters create any kind of confusion. But, merely running just one line of code, doesn't solve the purpose. Neither just looking at R^2 or MSE values. Regression tells much more than that!

In R, regression analysis return 4 plots using `plot(model_name)` function. Each of the plot provides significant information or rather an interesting story about the data. Sadly, many of the beginners either fail to decipher the information or don't care about what these plots say. Once you understand these plots, you'd be able to bring significant improvement in your regression model.

For model improvement, you also need to understand regression assumptions and ways to fix them when they get violated.

In this article, I've explained the important regression assumptions and plots (with fixes and solutions) to help you understand the regression concept in further detail. As said above, with this knowledge you can bring drastic improvements in your models.

Note: To understand these plots, you must know basics of regression analysis. If you are completely new to it, you can start here (<https://www.analyticsvidhya.com/blog/2015/10/regression-python-beginners/>). Then, proceed with this article.



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Assumptions in Regression

Regression is a parametric approach. 'Parametric' means it makes assumptions about data for the purpose of analysis. Due to its parametric side, regression is restrictive in nature. It fails to deliver good results with data sets which doesn't fulfill its assumptions. Therefore, for a successful regression analysis, it's essential to validate these assumptions.

So, how would you check (validate) if a data set follows all regression assumptions? You check it using the regression plots (explained below) along with some statistical test.

Let's look at the important assumptions in regression analysis:

1. There should be a **linear and additive relationship between dependent (response) variable and independent (predictor) variable(s)**. A linear relationship suggests that a change in response Y due to one unit change in X^1 is constant, regardless of the value of X^1 . An additive relationship suggests that the effect of X^1 on Y is independent of other variables.
2. **There should be no correlation between the residual (error) terms. Absence of this phenomenon is known as Autocorrelation.**
3. The independent variables should not be correlated. Absence of this phenomenon is known as **multicollinearity**.
4. The error terms must **have constant variance**. This phenomenon is known as homoskedasticity. The presence of non-constant variance is referred to heteroskedasticity.
5. The error terms must be **normally distributed**.

What if these assumptions get violated ?

Let's dive into specific assumptions and learn about their outcomes (if violated):

1. Linear and Additive: If you fit a linear model to a non-linear, non-additive data set, the regression algorithm would fail to capture the trend mathematically, thus resulting in an inefficient model. Also, this will result in erroneous predictions on an unseen data set.

How to check: Look for **residual vs fitted value plots** (explained below). Also, you can include polynomial terms (X , X^2 , X^3) in your model to capture the non-linear effect.

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2. Autocorrelation: The presence of correlation in error terms drastically reduces model's accuracy. This usually occurs in time series models where the next instant is dependent on previous instant. If the error terms are correlated, the estimated standard errors tend to underestimate the true standard error.

If this happens, it causes confidence intervals and prediction intervals to be narrower. Narrower confidence interval means that a 95% confidence interval would have lesser probability than 0.95 that it would contain the actual value of coefficients. Let's understand narrow prediction intervals with an example:

For example, the least square coefficient of X^1 is 15.02 and its standard error is 2.08 (without autocorrelation). But in presence of autocorrelation, the standard error reduces to 1.20. As a result, the prediction interval narrows down to (13.82, 16.22) from (12.94, 17.10).

Also, lower standard errors would cause the associated p-values to be lower than actual. This will make us incorrectly conclude a parameter to be statistically significant.

How to check: Look for **Durbin – Watson (DW) statistic**. It must lie between 0 and 4. If $DW = 2$, implies no autocorrelation, $0 < DW < 2$ implies positive autocorrelation while $2 < DW < 4$ indicates negative autocorrelation. Also, you can see residual vs time plot and look for the seasonal or correlated pattern in residual values.

3. Multicollinearity: This phenomenon exists when the independent variables are found to be moderately or highly correlated. In a model with correlated variables, it becomes a tough task to figure out the true relationship of a predictors with response variable. In other words, it becomes difficult to find out which variable is actually contributing to predict the response variable.

Another point, with presence of correlated predictors, the standard errors tend to increase. And, with large standard errors, the confidence interval becomes wider leading to less precise estimates of slope parameters.

Also, when predictors are correlated, the estimated regression coefficient of a correlated variable depends on which other predictors are available in the model. If this happens, you'll end up with an incorrect conclusion that a variable strongly / weakly affects target variable. Since, even if you drop one correlated variable from the model, its estimated regression coefficients would change. That's not good!

How to check: You can use scatter plot to visualize correlation effect among variables. Also, you can also use **VIF factor**. VIF value ≤ 4 suggests no multicollinearity whereas a value of ≥ 10 implies serious multicollinearity.

Above all, a correlation table should also solve the purpose.

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4. Heteroskedasticity: The presence of non-constant variance in the error terms results in heteroskedasticity. Generally, non-constant variance arises in presence of outliers or extreme leverage values. Look like, these values get too much weight, thereby disproportionately influences the model's performance. When this phenomenon occurs, the confidence interval for out of sample prediction tends to be unrealistically wide or narrow.

How to check: You can look at **residual vs fitted values plot**. If heteroskedasticity exists, the plot would exhibit a **funnel shape pattern** (shown in next section). Also, you can use Breusch-Pagan / Cook – Weisberg test or White general test to detect this phenomenon.

5. Normal Distribution of error terms: If the error terms are non- normally distributed, confidence intervals may become too wide or narrow. Once confidence interval becomes unstable, it leads to difficulty in estimating coefficients based on minimization of least squares. Presence of non – normal distribution suggests that there are a few unusual data points which must be studied closely to make a better model.

How to check: You can look at **QQ plot** (shown below). You can also perform statistical tests of normality such as Kolmogorov-Smirnov test, Shapiro-Wilk test.

Interpretation of Regression Plots

Until here, we've learnt about the important regression assumptions and the methods to undertake, if those assumptions get violated.

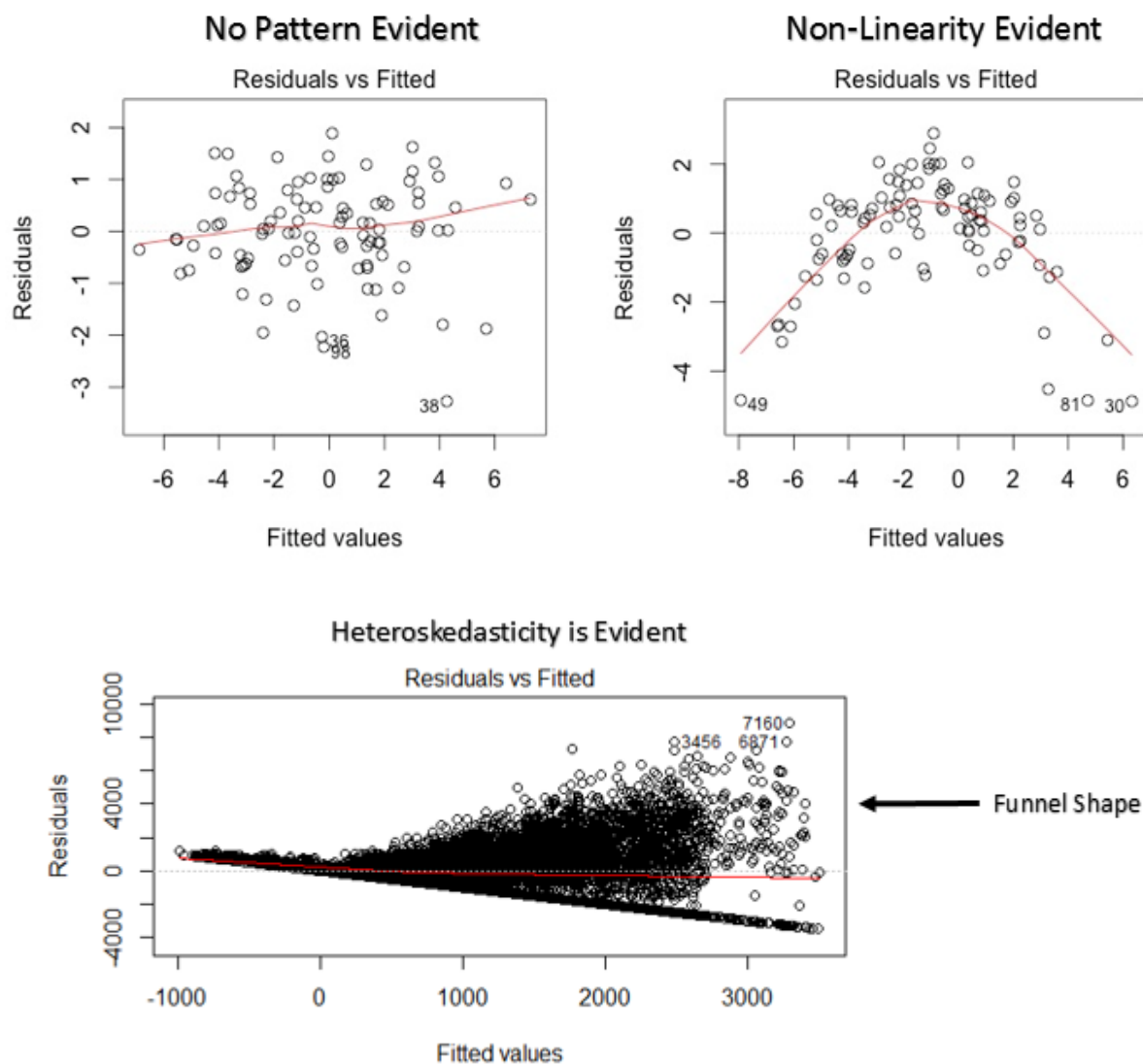
But that's not the end. Now, you should know the solutions also to tackle the violation of these assumptions. In this section, I've explained the 4 regression plots along with the methods to overcome limitations on assumptions.

1. Residual vs Fitted Values

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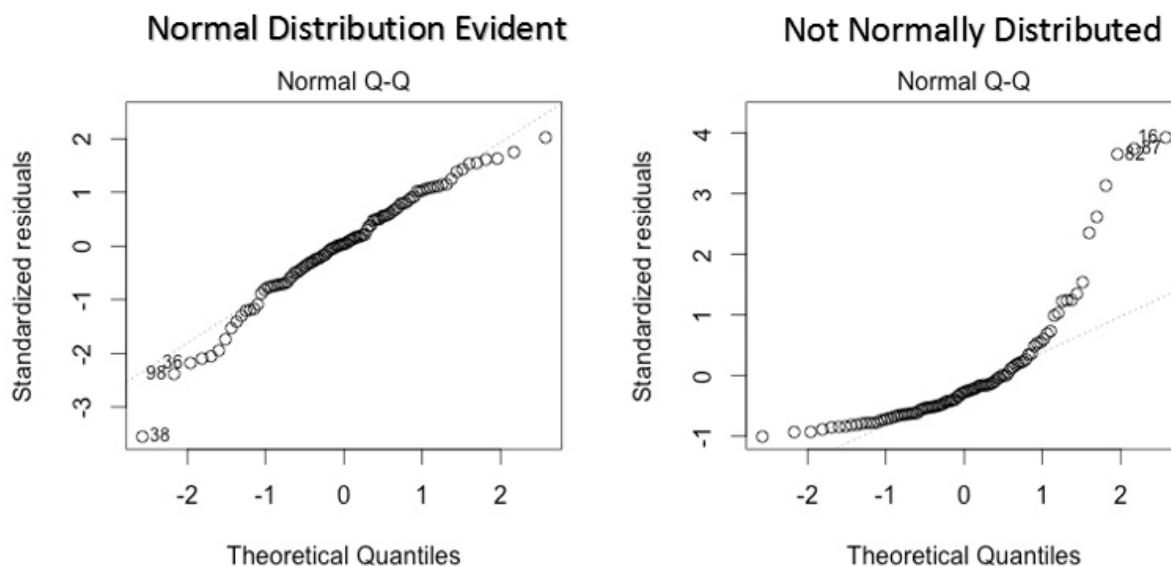
This scatter plot shows the distribution of residuals (errors) vs fitted values (predicted values). It is one of the most important plot which everyone must learn. It reveals various useful insights including outliers. The outliers in this plot are labeled by their observation number which make them easy to detect.

There are two major things which you should learn:

1. If there exist any pattern (may be, a parabolic shape) in this plot, consider it as signs of non-linearity in the data. It means that the model doesn't capture non-linear effects.
2. If a funnel shape is evident in the plot, consider it as the signs of non constant variance i.e. heteroskedasticity.

Solution: To overcome this issue of non-linearity, you can do a non-linear transformation of predictor or the response variable. To overcome heteroskedasticity, you can use the log transformation of the response variable. By using Analytics Vidya, you can transform the dependent variable (https://www.analyticsvidya.com/2016/07/10-ways-to-transform-the-response-variable/). Also, you can use the weighted least square method to tackle heteroskedasticity. (https://www.analyticsvidya.com/terms/). Accept

2. Normal Q-Q Plot



This q-q or quantile-quantile is a scatter plot which helps us validate the assumption of normal distribution in a data set. Using this plot we can infer if the data comes from a normal distribution. If yes, the plot would show fairly straight line. Absence of normality in the errors can be seen with deviation in the straight line.

If you are wondering what is a 'quantile', here's a simple definition: Think of quantiles as points in your data below which a certain proportion of data falls. Quantile is often referred to as percentiles. For example: when we say the value of 50th percentile is 120, it means half of the data lies below 120.

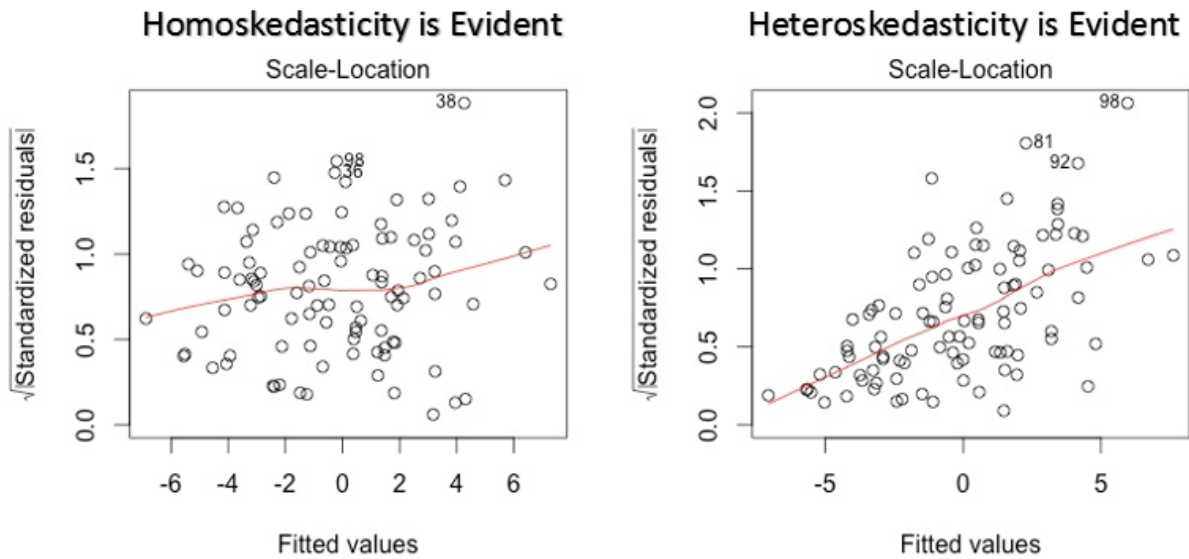
Solution: If the errors are not normally distributed, non – linear transformation of the variables (response or predictors) can bring improvement in the model.

3. Scale Location Plot

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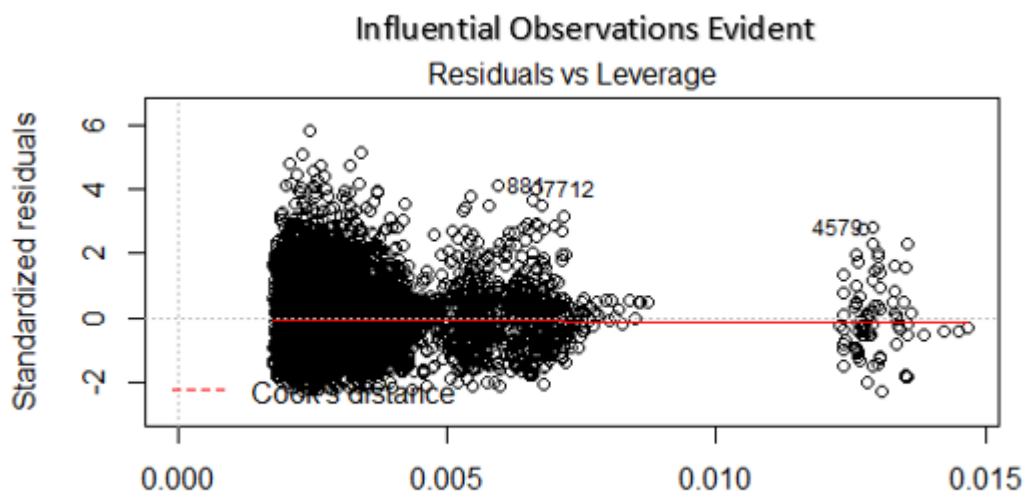
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This plot is also used to detect homoskedasticity (assumption of equal variance). It shows how the residual are spread along the range of predictors. It's similar to residual vs fitted value plot except it uses standardized residual values. Ideally, there should be no discernible pattern in the plot. This would imply that errors are normally distributed. But, in case, if the plot shows any discernible pattern (probably a funnel shape), it would imply non-normal distribution of errors.

Solution: Follow the solution for heteroskedasticity given in plot 1.

4. Residuals vs Leverage Plot



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It is also known as Cook's Distance plot. Cook's distance attempts to identify the points which have more influence than other points. Such influential points tends to have a sizable impact of the regression line. In other words, adding or removing such points from the model can completely change the model statistics.

But, can these influential observations be treated as outliers? This question can only be answered after looking at the data. Therefore, in this plot, the large values marked by cook's distance might require further investigation.

Solution: For influential observations which are nothing but outliers, if not many, you can remove those rows. Alternatively, you can scale down the outlier observation with maximum value in data or else treat those values as missing values.

Case Study: [How I improved my regression model using log transformation](https://www.analyticsvidhya.com/blog/2016/02/complete-tutorial-learn-data-science-scratch/#five)

(<https://www.analyticsvidhya.com/blog/2016/02/complete-tutorial-learn-data-science-scratch/#five>).

End Notes

You can leverage the true power of regression analysis by applying the solutions described above. Implementing these fixes in R is fairly easy. If you want to know about any specific fix in R, you can drop a comment, I'd be happy to help you with answers.

My motive of this article was to help you gain the underlying knowledge and insights of regression assumptions and plots. This way, you would have more control on your analysis and would be able to modify the analysis as per your requirement.

Did you find this article useful ? Have you used these fixes in improving model's performance? Share your experience / suggestions in the comments.

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For linear classifiers, do larger coefficients imply more important features?

I'm a software engineer working on machine learning. From my understanding, **linear regression** (such as OLS) and **linear classification** (such as logistic regression and SVM) make a prediction based on an inner product between trained coefficients $\vec{w}$ and feature variables $\vec{x}$:

$$\hat{y} = f(\vec{w} \cdot \vec{x}) = f\left(\sum_i w_i x_i\right)$$

My question is: After the model has been trained (that is, after the coefficients w_i have been computed), is it the case that the coefficients will be larger for feature variables that are more important for the model to predict more accurately?

In other words, I am asking whether the relative magnitudes of the coefficients can be used for **feature selection** by just ordering the variables by coefficient value and then selecting the features with the highest coefficients? If this approach is valid, then why is it not mentioned for feature selection (along with wrapper and filter methods, etc.).

The reason I ask this is because I came across a discussion on **L1 vs. L2 regularization**. There is a blurb that says:

Built-in feature selection is frequently mentioned as a useful property of the L1-norm, which the L2-norm does not. This is actually a result of the L1-norm, which tends to produce sparse coefficients (explained below). Suppose the model has 100 coefficients but only 10 of them have non-zero coefficients, this is effectively saying that "the other 90 predictors are useless in predicting the target values".

Reading between the lines, I would guess that if a coefficient is close to 0, then the feature variable with that coefficient must have little predictive power.

EDIT: I am also applying z-scaling to my numeric variables.

regression machine-learning classification feature-selection linear-model

edited Mar 21 '16 at 18:40

asked Mar 17 '16 at 17:12

 **stackoverflowuser2010**
1,080 3 17 30

- 1  Note that the code underlying LASSO (L1-norm) and ridge regression (L2-norm) analyses should pre-scale the predictor variables prior to analysis, even if the code then transforms the coefficients back into the original variable scales. Those who use code that doesn't pre-scale end up with the problems noted in the answer from @josliber whether they are doing OLS, LASSO, or ridge. – **EdM** Mar 21 '16 at 18:30 
- 3  I think something worth mentioning is, when you reflect on what are trying to express by the phrase "then the feature variable with that coefficient must have little predictive power", can you precisely lay out what that really means? I've found though experience that the concept of "predictive power" of an individual variable in a multivariate model has no generally agreed upon conceptual foundation. – **Matthew Drury** Mar 21 '16 at 18:55 
- 4  I think the error in that kind of thinking is that you are probably *not* confined to producing a one variable model. If you are, and you want to provide a model with the best accuracy, they sure, that is a reasonable thing to do. If you are not, i.e. if you are going to produce a multivariate model, then, as @EdM answers, the concept of variable importance is very, very slippery, and lacks a firm conceptual foundation. It is not at all obvious that predictive power in a univariate model should be seen as relevant in a multivariate setting. – **Matthew Drury** Mar 21 '16 at 20:36
- 1  @MatthewDrury: I'm not sure why you are making a big deal out of multi-features. There is a whole field of "feature selection" (e.g. wrapper methods) that exists; are you suggesting that this field lacks a firm conceptual foundation? – **stackoverflowuser2010** Mar 22 '16 at 17:54
- 1  @stackoverflowuser2010 Yah, I'm probably an outlier in my opinion here, but that would be a somewhat accurate description of my perspective. – **Matthew Drury** Mar 22 '16 at 21:21

3 Answers

Not at all. The magnitude of the coefficients depends directly on the scales selected for the variables, which is a somewhat arbitrary modeling decision.

To see this, consider a linear regression model predicting the petal width of an iris (in centimeters)

given its petal length (in centimeters):

```
summary(lm(Petal.Width~Petal.Length, data=iris))
# Call:
# lm(formula = Petal.Width ~ Petal.Length, data = iris)
#
# Residuals:
#      Min       1Q   Median       3Q      Max
# -0.56515 -0.12358 -0.01898  0.13288  0.64272
#
# Coefficients:
#              Estimate Std. Error t value Pr(>|t|)
# (Intercept)  -0.363076   0.039762  -9.131  4.7e-16 ***
# Petal.Length   0.415755   0.009582  43.387  < 2e-16 ***
# ---
# Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#
# Residual standard error: 0.2065 on 148 degrees of freedom
# Multiple R-squared:  0.9271, Adjusted R-squared:  0.9266
# F-statistic: 1882 on 1 and 148 DF, p-value: < 2.2e-16
```

Our model achieves an adjusted R^2 value of 0.9266 and assigns coefficient value 0.415755 to the Petal.Length variable.

However, the choice to define Petal.Length in centimeters was quite arbitrary, and we could have instead defined the variable in meters:

```
iris$Petal.Length.Meters <- iris$Petal.Length / 100
summary(lm(Petal.Width~Petal.Length.Meters, data=iris))
# Call:
# lm(formula = Petal.Width ~ Petal.Length.Meters, data = iris)
#
# Residuals:
#      Min       1Q   Median       3Q      Max
# -0.56515 -0.12358 -0.01898  0.13288  0.64272
#
# Coefficients:
#              Estimate Std. Error t value Pr(>|t|)
# (Intercept)    -0.36308    0.03976  -9.131  4.7e-16 ***
# Petal.Length.Meters 41.57554    0.95824  43.387  < 2e-16 ***
# ---
# Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#
# Residual standard error: 0.2065 on 148 degrees of freedom
# Multiple R-squared:  0.9271, Adjusted R-squared:  0.9266
# F-statistic: 1882 on 1 and 148 DF, p-value: < 2.2e-16
```

Of course, this doesn't really affect the fitted model in any way -- we simply assigned a 100x larger coefficient to Petal.Length.Meters (41.57554) than we did to Petal.Length (0.415755). All other properties of the model (adjusted R^2 , t-statistics, p-values, etc.) are identical.

Generally when fitting regularized linear models one will first normalize variables (for instance, to have mean 0 and unit variance) to avoid favoring some variables over others based on the selected scales.

Assuming Normalized Data

Even if you had normalized all variables, variables with higher coefficients may still not be as useful in predictions because the independent variables are rarely set (have low variance). As an example, consider a dataset with dependent variable Z and independent variables X and Y taking binary values

```
set.seed(144)
dat <- data.frame(X=rep(c(0, 1), each=50000),
                  Y=rep(c(0, 1), c(1000, 99000)))
dat$Z <- dat$X + 2*dat$Y + rnorm(100000)
```

By construction, the coefficient for Y is roughly twice as large as the coefficient for X when both are used to predict Z via linear regression:

```
summary(lm(Z~X+Y, data=dat))
# Call:
# lm(formula = Z ~ X + Y, data = dat)
#
# Residuals:
#      Min       1Q   Median       3Q      Max
# -4.4991 -0.6749 -0.0056  0.6723  4.7342
#
# Coefficients:
#              Estimate Std. Error t value Pr(>|t|)
```

```
# (Intercept) -0.094793  0.031598  -3.00  0.0027 **
# X           0.999435  0.006352 157.35 <2e-16 ***
# Y           2.099410  0.031919  65.77  <2e-16 ***
# ---
# Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#
# Residual standard error: 0.9992 on 99997 degrees of freedom
# Multiple R-squared:  0.2394, Adjusted R-squared:  0.2394
# F-statistic: 1.574e+04 on 2 and 99997 DF, p-value: < 2.2e-16
```

Still, X explains more of the variance in Z than Y (the linear regression model predicting Z with X has R^2 value 0.2065, while the linear regression model predicting Z with Y has R^2 value 0.0511):

```
summary(lm(Z~X, data=dat))
# Call:
# lm(formula = Z ~ X, data = dat)
#
# Residuals:
#    Min       1Q   Median       3Q      Max
# -5.2587 -0.6759  0.0038  0.6842  4.7342
#
# Coefficients:
#              Estimate Std. Error t value Pr(>|t|)
# (Intercept)  1.962629   0.004564   430.0 <2e-16 ***
# X            1.041424   0.006455   161.3 <2e-16 ***
# ---
# Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#
# Residual standard error: 1.021 on 99998 degrees of freedom
# Multiple R-squared:  0.2065, Adjusted R-squared:  0.2065
# F-statistic: 2.603e+04 on 1 and 99998 DF, p-value: < 2.2e-16
```

versus:

```
summary(lm(Z~Y, data=dat))
# Call:
# lm(formula = Z ~ Y, data = dat)
#
# Residuals:
#    Min       1Q   Median       3Q      Max
# -5.0038 -0.7638 -0.0007  0.7610  5.2288
#
# Coefficients:
#              Estimate Std. Error t value Pr(>|t|)
# (Intercept) -0.09479   0.03529  -2.686  0.00724 **
# Y            2.60418    0.03547  73.416 < 2e-16 ***
# ---
# Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#
# Residual standard error: 1.116 on 99998 degrees of freedom
# Multiple R-squared:  0.05114, Adjusted R-squared:  0.05113
# F-statistic: 5390 on 1 and 99998 DF, p-value: < 2.2e-16
```

The Case of Multi-Collinearity

A third case where large coefficient values may be deceiving would be in the case of significant multi-collinearity between variables. As an example, consider a dataset where X and Y are highly correlated but W is not highly correlated to the other two; we are trying to predict Z:

```
set.seed(144)
dat <- data.frame(W=rnorm(100000),
                  X=rnorm(100000))
dat$Y <- dat$X + rnorm(100000, 0, 0.001)
dat$Z <- 2*dat$W+10*dat$X-11*dat$Y + rnorm(100000)
cor(dat)
#           W           X           Y           Z
# W 1.000000e+00  5.191809e-05  5.200434e-05  0.8161636
# X 5.191809e-05  1.000000e+00  9.999995e-01 -0.4079183
# Y 5.200434e-05  9.999995e-01  1.000000e+00 -0.4079246
# Z 8.161636e-01 -4.079183e-01 -4.079246e-01  1.0000000
```

These variables pretty much have the same mean (0) and variance (~1), and linear regression assigns much higher coefficient values (in absolute value) to X (roughly 15) and Y (roughly -16) than it does to W (roughly 2):

```
summary(lm(Z~W+X+Y, data=dat))
# Call:
# lm(formula = Z ~ W + X + Y, data = dat)
#
# Residuals:
```

```
#      Min      1Q  Median      3Q      Max
# -4.1886 -0.6760  0.0026  0.6679  4.2232
#
# Coefficients:
#              Estimate Std. Error t value Pr(>|t|)
# (Intercept)  1.831e-04  3.170e-03   0.058   0.954
# W            2.001e+00  3.172e-03 630.811 < 2e-16 ***
# X            1.509e+01  3.177e+00  4.748 2.05e-06 ***
# Y           -1.609e+01  3.177e+00 -5.063 4.13e-07 ***
# ---
# Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
#
# Residual standard error: 1.002 on 99996 degrees of freedom
# Multiple R-squared:  0.8326, Adjusted R-squared:  0.8326
# F-statistic: 1.658e+05 on 3 and 99996 DF, p-value: < 2.2e-16
```

Still, among the three variables in the model W is the most important: If you remove W from the full model, the R^2 drops from 0.833 to 0.166, while if you drop X or Y the R^2 is virtually unchanged.

edited Mar 22 '16 at 14:35

answered Mar 21 '16 at 18:24



amoeba

53k 13 182 238



josliber

2,699 15 36

- 1 (+1) I think this is a good answer, and this point must be made. On the other hand, I do believe there is a lot to say even when the variables have been standardized (and are hence, unit-less), so there's room for a few more answers. – [Matthew Drury](#) Mar 21 '16 at 18:31
- Thanks for the answer. While your writing is long, it's not very thorough because you're drawing conclusions from small synthetic data. Also, R^2 is specific to linear regression. I believe the more appropriate error metric is RMSE, or accuracy / F1 for classification problems. – [stackoverflowuser2010](#) Mar 29 '16 at 2:36
- 1 BUT if you scale the data in the last example, you have that the only significant variable is W – [marcodena](#) Aug 16 '16 at 17:40
- @marcodena thank you for pointing this out – [jaranda](#) Sep 28 '17 at 11:57

"Feature importance" is a very slippery concept even when all predictors have been adjusted to a common scale (which in itself is a non-trivial problem in many practical applications involving categorical variables or skewed distributions). So if you avoid the scaling problems indicated in the answer by [@josliber](#) or the low-predictor-variance issue raised by [@dsaxton](#), you still have additional problems.

For example, a more useful measure of feature importance may be the ratio of its coefficient to the estimated error of its coefficient. A high coefficient with a large estimated error would not necessarily be helpful in predictions. So coefficient magnitude alone, even in the pre-scaled situation, is not a good guide to "importance."

Nevertheless, a predictor may be important even if its coefficient's ratio of magnitude to error is low (i.e., it is not "statistically significant"). Linear models provide the ability to take multiple predictor variables into account simultaneously, so including a "non-significant" predictor in a model can improve the overall performance provided by the combined collection of predictors.

Furthermore, attempts to select "important" predictor variables tend to be highly dependent on the particular data sample and often do not extend well to further samples, particularly if the variables are correlated. You can see this for yourself by repeating feature selection on multiple bootstrap samples of the same data set. Frank Harrell, in [this answer](#) shows how to use his `rms` package in R for ranking feature importance, and notes in [this answer](#) how to use the bootstrap to get confidence intervals for the ranks. Bootstrapping can serve as a caution to those who put too much importance in "feature importance."

[This question](#) from nearly 3 years ago, pointed out by [@amoeba](#), also goes into extensive detail about difficulties with feature importance in multiple regression models.



How to Identify the Most Important Predictor Variables in Regression Models

Minitab Blog Editor · 07 September, 2016

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You've performed multiple linear regression and have settled on a model which contains several predictor variables that are statistically significant. At this point, it's common to ask, "Which variable is most important?"

This question is more complicated than it first appears. For one thing, how you define "most important" often depends on your subject area and goals. For another, how you collect and measure your sample data can influence the apparent importance of each variable.

With these issues in mind, I'll help you answer this question. I'll start by showing you statistics that *don't* answer the question about importance, which may surprise you. Then, I'll move on to both statistical and non-statistical methods for determining which variables are the most important in regression models.



Don't Compare Regular Regression Coefficients to Determine Variable Importance

Regular [regression coefficients](#) describe the relationship between each [predictor variable](#) and the [response](#). The coefficient value represents the mean change in the response given a one-unit increase in the predictor. Consequently, it's easy to think that variables with larger coefficients are more important because they represent a larger change in the response.

However, the units vary between the different types of variables, which makes it impossible to compare them directly. For example, the meaning of a one-unit change is very different if

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you're talking about temperature, weight, or chemical concentration.

This problem is further complicated by the fact that **there are different units within each type of measurement**. For example, weight can be measured in grams and kilograms. If you fit models for the same data set using grams in one model and kilograms in another, the coefficient for weight changes by a factor of a thousand even though the underlying fit of the model remains unchanged. The coefficient value changes greatly while the importance of the variable remains constant.

Takeaway: Larger coefficients don't necessarily identify more important predictor variables.

Don't Compare P-values to Determine Variable Importance

The coefficient value doesn't indicate the importance a variable, but what about the variable's p-value? After all, we look for low p-values to help determine whether the variable should be included in the model in the first place.

P-value calculations incorporate a variety of properties, but a measure of importance is not among them. **A very low p-value can reflect properties other than importance, such as a very precise estimate and a large sample size.**

Effects that are trivial in the real world can have very low p-values. A statistically significant result may not be practically significant.

Takeaway: Low p-values don't necessarily identify predictor variables that are practically important.

Do Compare These Statistics To Help Determine Variable Importance

We ruled out a couple of the more obvious statistics that can't assess the importance of variables. Fortunately, there are several statistics that can help us determine which predictor variables are most important in regression models. These statistics might not agree because the manner in which each one defines "most important" is a bit different.

Standardized regression coefficients

I explained how regular regression coefficients use different scales and you can't compare them directly. However, if you standardize the regression coefficients so they're based on the same scale, you *can* compare them.

To obtain standardized coefficients, standardize the values for all of your continuous predictors. In [Minitab](#), you can do this easily by clicking the **Coding** button in the main Regression dialog. Under **Standardize continuous predictors**, choose *Subtract the mean, then divide by the standard deviation*.

After you fit the regression model using your standardized predictors, look at the *coded*

coefficients, which are the standardized coefficients. This coding puts the different predictors on the same scale and allows you to compare their coefficients directly. Standardized coefficients represent the mean change in the response given a one standard deviation change in the predictor.

Takeaway: Look for the predictor variable with the largest absolute value for the standardized coefficient.

Change in R-squared when the variable is added to the model last

Multiple regression in Minitab's Assistant menu includes a neat analysis. It calculates the increase in R-squared that each variable produces when it is added to a model that already contains all of the other variables.

Because the change in R-squared analysis treats each variable as the last one entered into the model, the change represents the percentage of the variance a variable explains that the other variables in the model cannot explain. In other words, this change in R-squared represents the amount of *unique* variance that each variable explains above and beyond the other variables in the model.

Takeaway: Look for the predictor variable that is associated with the greatest increase in R-squared.

An Example of Using Statistics to Identify the Most Important Variables in a Regression Model

The example output below shows a regression model that has three predictors. The text output is produced by the regular regression analysis in Minitab. I've standardized the continuous predictors using the **Coding** dialog so we can see the standardized coefficients, which are labeled as coded coefficients. You can find this analysis in the Minitab menu: **Stat > Regression > Regression > Fit Regression Model**.

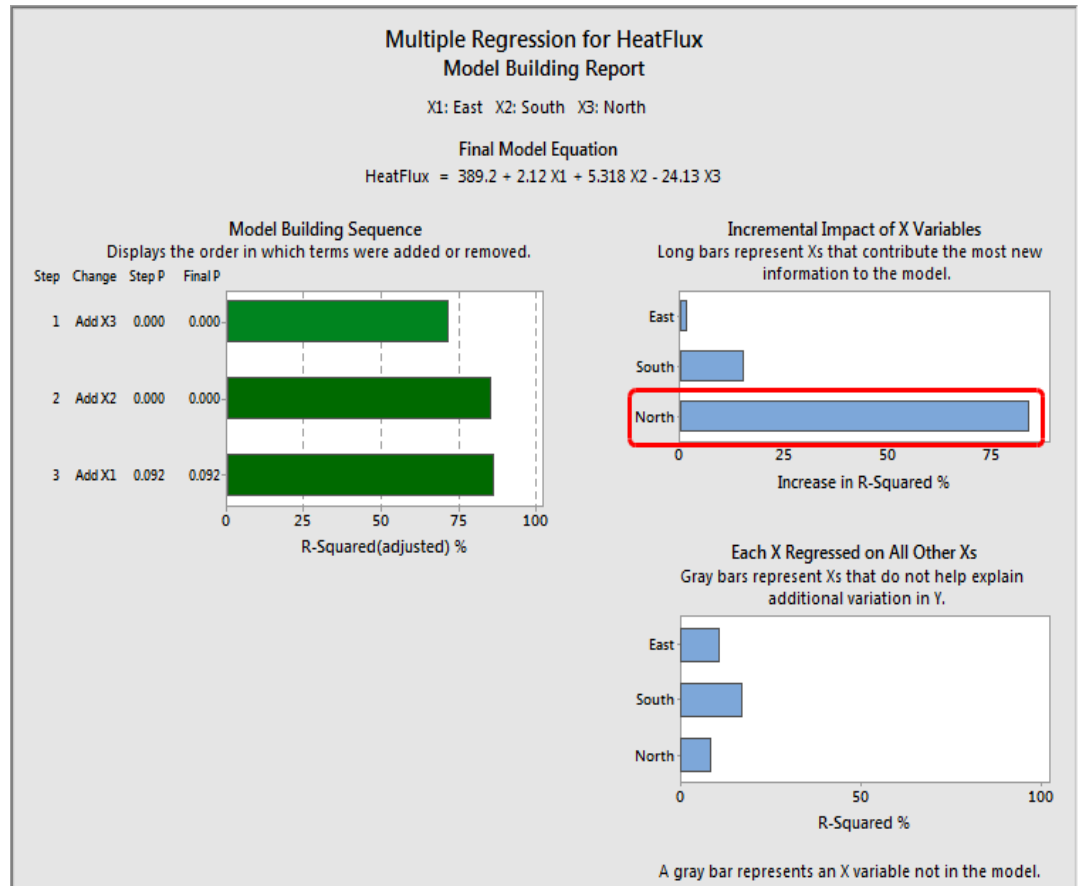
The report with the graphs is produced by Multiple Regression in the Assistant menu. You can find this analysis in the Minitab menu: **Assistant > Regression > Multiple Regression**.

Coded Coefficients

Term	Coef	SE Coef	T-Value	P-Value	VIF
Constant	249.64	1.60	156.36	0.000	
East	3.01	1.72	1.75	0.092	1.12
South	9.85	1.78	5.52	0.000	1.21
North	-21.92	1.70	-12.92	0.000	1.09

Regression Equation in Uncoded Units

HeatFlux = 389.2 + 2.12 East + 5.318 South - 24.13 North



The standardized coefficients show that North has the standardized coefficient with the largest absolute value, followed by South and East. The Incremental Impact graph shows that North explains the greatest amount of the unique variance, followed by South and East. For our example, both statistics suggest that North is the most important variable in the regression model.

Caveats for Using Statistics to Identify Important Variables

Statistical measures can show the relative importance of the different predictor variables. However, these measures can't determine whether the variables are important in a practical sense. To determine practical importance, you'll need to use your subject area knowledge.

How you collect and measure your sample can bias the apparent importance of the variables in your sample compared to their true importance in the population.

If you randomly sample your observations, the variability of the predictor values in your sample likely reflects the variability in the population. In this case, the standardized coefficients and the change in R-squared values are likely to reflect their population values.

However, if you select a restricted range of predictor values for your sample, both statistics tend to underestimate the importance of that predictor. Conversely, if the sample variability for a predictor is greater than the variability in the population, the statistics tend to

overestimate the importance of that predictor.

Also, consider the accuracy and precision of the measurements for your predictors because this can affect their apparent importance. For example, lower-quality measurements can cause a variable to appear less predictive than it truly is.

If your goal is to change the response mean, you should be confident that causal relationships exist between the predictors and the response rather than just a correlation. If there is an observed [correlation but no causation](#), intentional changes in the predictor values won't necessarily produce the desired change in the response regardless of the statistical measures of importance.

To determine that there is a causal relationship, you typically need to perform a [designed experiment rather than an observational study](#).

Non-Statistical Considerations for Identifying Important Variables

How you define "most important" often depends on your goals and subject area. While statistics can help you identify the most important variables in a regression model, applying subject area expertise to all aspects of statistical analysis is crucial. Real world issues are likely to influence which variable you identify as the most important in a regression model.

For example, if your goal is to change predictor values in order to change the response, use your expertise to determine which variables are the most feasible to change. There may be variables that are harder, or more expensive, to change. Some variables may be impossible to change. Sometimes a large change in one variable may be more practical than a small change in another variable.

"Most important" is a subjective, context sensitive characteristic. You can use statistics to help identify candidates for the most important variable in a regression model, but you'll likely need to use your subject area expertise as well.

If you're just learning about regression, read my [regression tutorial](#)!

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